

Phylogenetic analysis of HIV-1 *pol* sequences from 3 distinct regions: the United States, Mozambique, and Thailand

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Abstract

Advancing treatment and research relating to HIV/AIDS is a global effort. But with a virus that impacts such a vast range of the world, variation between regional subtypes is common, emphasizing the value of finding treatments that target similar or conserved regions of the genome. The *pol* gene is a region that is known to encode for many resistance factors and that is conserved enough between subtypes for the use of universal primers. This study analyzes an alignment of 5 random HIV-1 *pol* BLAST sequences from 3 different countries (United States, Mozambique, Thailand) using the methods of neighbor-joining, maximum likelihood, and Bayesian inference to derive phylogenetic trees. Consistent clustering by country was seen across all built trees, showing differences between HIV-1 *pol* sequences regionally. Variation was also seen within each country, demonstrated by the unique patterns of internal substructures. Although support values across these trees were highly variable and Effective Sample Size (ESS) values were low, in part due to the small sample size ($n=15$), valuable patterns were demonstrated that can be used to guide future work.

Introduction

Human immunodeficiency virus (HIV) is a pathogen with monumental global history and impacts. It is spread through specific contact with certain bodily fluids, such as through unprotected sex or via shared needles. It is important to also emphasize that HIV cannot be transmitted through casual contact, such as hugs or handshakes, as this perpetuates harmful and untrue stereotypes surrounding individuals with the virus. Once it enters the body, HIV targets and destroys CD4 immune cells, leading to a decreased ability to fight off infections (HIV.gov, 2023). If left untreated, the immune system deterioration caused by HIV can contribute to numerous complications, including a high risk for opportunistic infections, which is associated with HIV's eventual progression into acquired immunodeficiency syndrome (AIDS).

That being said, people with HIV who receive treatment can live normal, healthy lives. This is done through antiretroviral therapy (ART), which consists of medications that, when taken as prescribed, lowers an individual's viral load to an undetectable level (HIV.gov, 2023). HIV that is undetectable is untransmittable (UNAIDS, 2018), emphasizing just how crucial it is that ART remains effective and accessible. However, viruses evolve. Selective pressures can lead to the maintenance of favorable resistance traits in some viral strains, and ART advancement must be able to keep up.

The *pol* gene is a key sequence when it comes to studying and understanding resistance across the various HIV-1 subtypes. Not only is it a region that encodes for many of the traits associated with resistance (Blassel et al., 2021), but it has also been found to be generally conserved enough across subtypes for use of universal primers, making it a good region to study for treatment development (Nie et al., 2011). This is especially crucial considering how research tends to be concentrated around region-specific subtypes. For example, subtype B is dominant in North American and European regions and thus the primary target of research in these countries. However, subtype B only makes up around 12% of global HIV cases, with A and C dominating across much of the rest of the world (Nie et al., 2011). Because of this, research efforts and treatment development that is applicable across various subtypes is of high value when it comes to fighting HIV globally, including in areas which experience the impacts of the virus disproportionately.

Phylogenetics in research involving treatment resistance is especially useful because it allows genes and traits to be traced back to their evolutionary origins. This knowledge provides important insight when it comes to advancing our understanding of the sources of resistance mutations. This study aims to assess regional variation in *pol* gene phylogeny by processing random *pol* BLAST sequences from 3 different countries – the United States, Mozambique, and Thailand – and then developing trees that can be used to compare trends in mutation development.

Materials & Methods

Data Acquisition

Data used in this project was obtained from the Los Alamos National Laboratory HIV sequence database. Important parameters specified included: Sampling year (2018), virus (HIV-1), subtype (any subtype), genomic region (Pol CDS), and geographic region (North America/Africa/Asia). Five random BLAST *pol* sequences of each country's most prominent HIV-1 subtype were selected from the three chosen countries (United States, Mozambique, Thailand). While 15 distinct sequences were initially selected, it was later determined that 2 sequences from Mozambique were identical, resulting in 14 unique samples used across further analyses.

Country	Subtype	Accession Numbers				
United States	B	MK169417	MK169418	MK169422	MK169441	MK169431
Mozambique	C	OK649266	OK649268*	OK649270*	OK649287	OK649295
Thailand	CRF01_AE	ON903098	ON903078	ON863057	ON863022	ON863108

Table 1. HIV-1 *pol* BLAST sequences used: subtypes and accession numbers (Los Alamos National Laboratory, 2023). *Later analysis determined as identical sequences.

MAFFT for Multiple Sequence Alignment

The first step in analyzing the 15 sequences was to create an alignment. MAFFT is a multiple sequence alignment program that works via two main techniques – identification of homologous regions and use of an improved, more efficient scoring system. Assumptions when using MAFFT include that the chosen sequence regions are homologous and closely related, and that alignments are based on algorithms like Needleman-Wunsch for computing alignment without rearrangement. Some limitations include how the Fast Fourier Transform algorithm is not efficient for more distantly related sequences, as well as increased CPU times for less conserved regions (Kato et al., 2002).

To test for quality control, code was run to trim the alignment. This removed any gaps that there may have been. After comparing trimmed and untrimmed IQ-TREE and MrBayes outputs, they were similar enough to justify not reworking the studies using the trimmed alignment.

Distance and Parsimony Trees

Distance-based trees are built by computation of pairwise genetic distances and hierarchical clustering algorithms, in this case Neighbor-Joining. A TN93 model was chosen, leading to the assumption of varying rates of transitions, transversions, and heterogeneous base frequencies. A limitation of this model is that the output is only 1 tree when in reality there could be multiple equally-possible options.

Parsimony-based trees are built upon the basis of using the smallest number of genetic sequence changes between each taxa. Assumptions include that the simplest scenario, with the least amount of genetic changes, is the most likely. This is more accurate when the genetic changes are less compared to something rapidly evolving and mutating. There are limitations to this, though, as assuming the simplest scenario doesn't account for large evolutionary events well. This model is also more simple and computer intensive than some of its counterparts. Because of there being no true outgroup, this tree was rooted arbitrarily for clarity.

Maximum Likelihood: RAxML

RAxML is a phylogenetics program used for maximum likelihood analysis of sequence data. It is known for generating high-scoring trees relative to other programs, and for doing so very quickly. The program assumes that inputs are from “classical GTR-derived models” (Kozlov et al., 2019), and that all sites have evolved independently. Limitations include being found as slightly less accurate than IQ-TREE. Additionally, in situations where phylogenetic signal is strong, IQ-TREE may require decreased replicate searches compared to RAxML (Kozlov et al., 2019). However, RAxML has shown to produce higher-scoring output trees on taxon-rich alignments when compared to IQ-TREE (Kozlov et al., 2019; Stamatakis, 2014).

This analysis was run on our dataset which contained 14 distinct sequences and 3094 alignment sites using the software RAxML-NG with GTR+G as the substitution model. It performed a maximum likelihood tree search with 20 distinct starting trees: 10 random and 10 parsimony. Bootstrap analysis

was run with 1000 replicates. While RAxML trees typically need to be rooted with an outgroup, this assessment was focusing more on clustering patterns of just HIV-1 *pol* sequences and therefore had no true outgroup.

Maximum Likelihood: IQ-TREE

IQ-TREE is a software used for phylogenetic inference by maximum likelihood (Minh et al., 2020; Nguyen et al., 2014). The software package includes model selection through ModelFinder, an effective search algorithm, and fast bootstrapping. Assumptions include treelikeness, constant frequency of nucleotides and amino acids through time, substitutions that are equally likely in both directions, and a substitution rate that is constant through time. Limitations include a potential need for parameter adjustment when phylogenetic signals are weak, results that are dependent on alignment quality, and the fact that extensive bootstrapping still presents uncertainty (Bui, 2021).

The aligned FASTA file was input in IQ-TREE, and model selection using ModelFinder identified TIM3+F+I as the best-fit substitution model for Bayesian Information Criterion. The model constructed a maximum likelihood tree constructed using 1000 ultrafast bootstrap replicates for branch support. It also brought attention to the two identical sequences denoted above.

Bayesian Inference: MrBayes

MrBayes is a software that uses the Markov Chain Monte Carlo (MCMC) method to make estimations for Bayesian inference modeling (Huelsenbeck & Ronquist, 2001). Bayesian inference begins with a prior probability that is continually updated with the addition of more data to produce a posterior probability. Assumptions made by this program include that data is homogenous and the correct substitution model was chosen. Additionally, concatenation methods implicitly assume that all gene loci share the same topology and branch lengths (Huelsenbeck and Ronquist, 2003). Limitations include how outputs are dependent on the quality of the input alignment and the correctness of chosen models.

After converting it from a FASTA file to a nexus file, the alignment was uploaded and run through MrBayes with the substitution model GTR+I. An MCMC analysis was run for 500,000 generations to give us an output of posterior probabilities. Tracer was used to assess convergence by looking at trace plots of posterior and likelihood values.

Bayesian Inference: BEAST

BEAST is a software package that uses MCMC for Bayesian analysis of molecular sequences. It is used to estimate evolutionary parameters. The aligned sequences are put into BEAUti, which generates the XML configuration file to input in BEAST. BEAST outputs log and trees files. Then, the log file (beast/beast_BEAUti2_analysis.log) is loaded into Tracer for analysis of posterior and likelihood trace plots and to assess parameters (BEAST2 Developers, 2026).

Distance Matrix

The 15 aligned sequences were used to create a distance matrix with the R package, ape. The output was saved as a CSV file to be easily accessible as a spreadsheet.

Results

Phylogenetic analysis of HIV-1 *pol* sequences from 3 regions with varying treatment contexts showed that the sequences clustered clearly by country. The 5 random BLAST sequences selected from the United States formed a uniform group, while the sequences from Thailand and Mozambique had subgroups within their clusters. Thailand's subgroups were more compact and defined in structure, while Mozambique's were more dispersed. These patterns were produced consistently across 3 trees built using different methods: RAxML and IQ-TREE (Figure 1) for maximum likelihood, and MrBayes (Figure 2) for Bayesian inference. This alignment also had a consistent high proportion of invariant sites across models of approximately 76%, demonstrating minimal variation in *pol* gene sequences across samples.

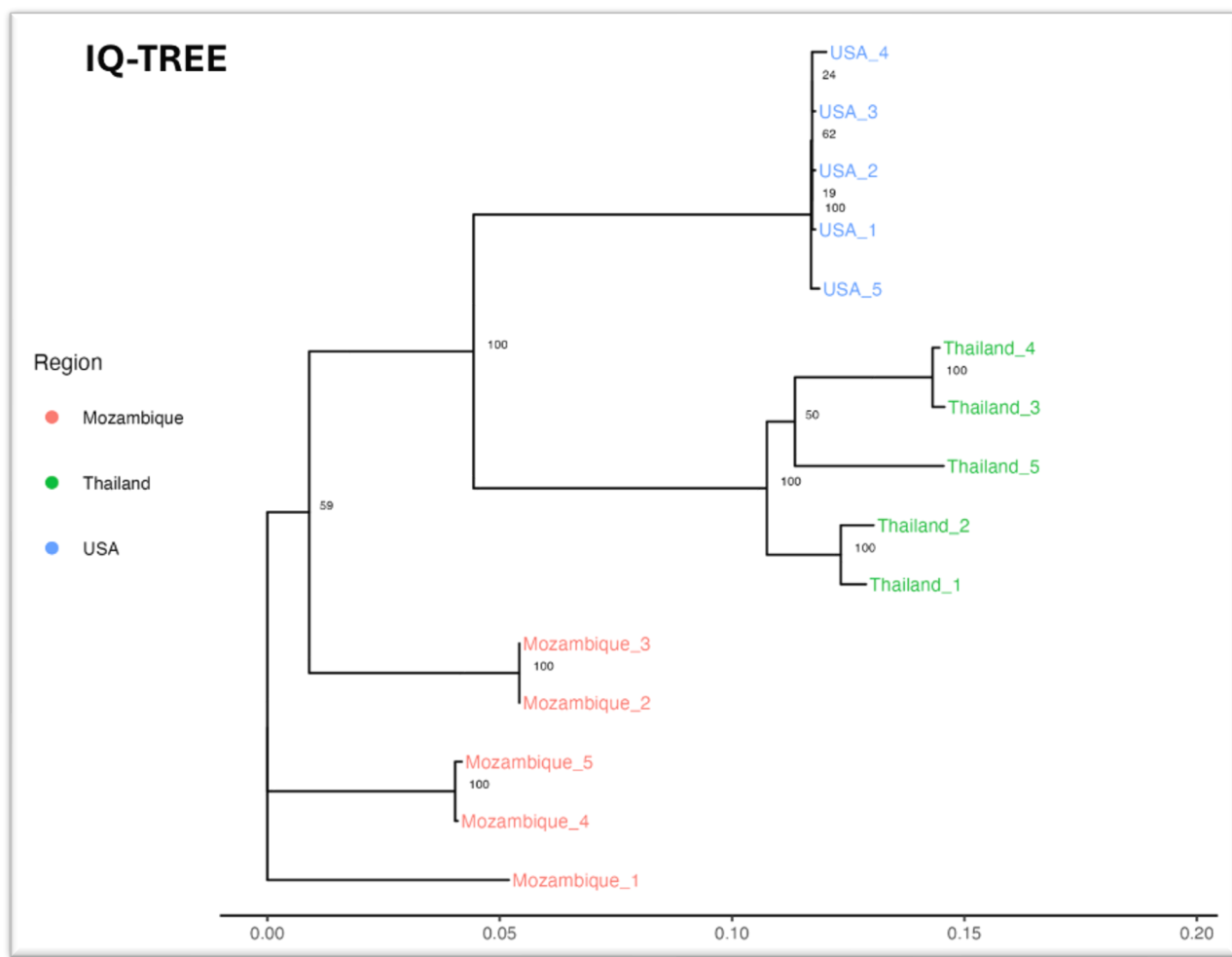


Figure 1. Maximum likelihood tree constructed with IQ-TREE. Numbers at the nodes indicate frequency of occurrence across 1000 ultrafast bootstrap replicates.

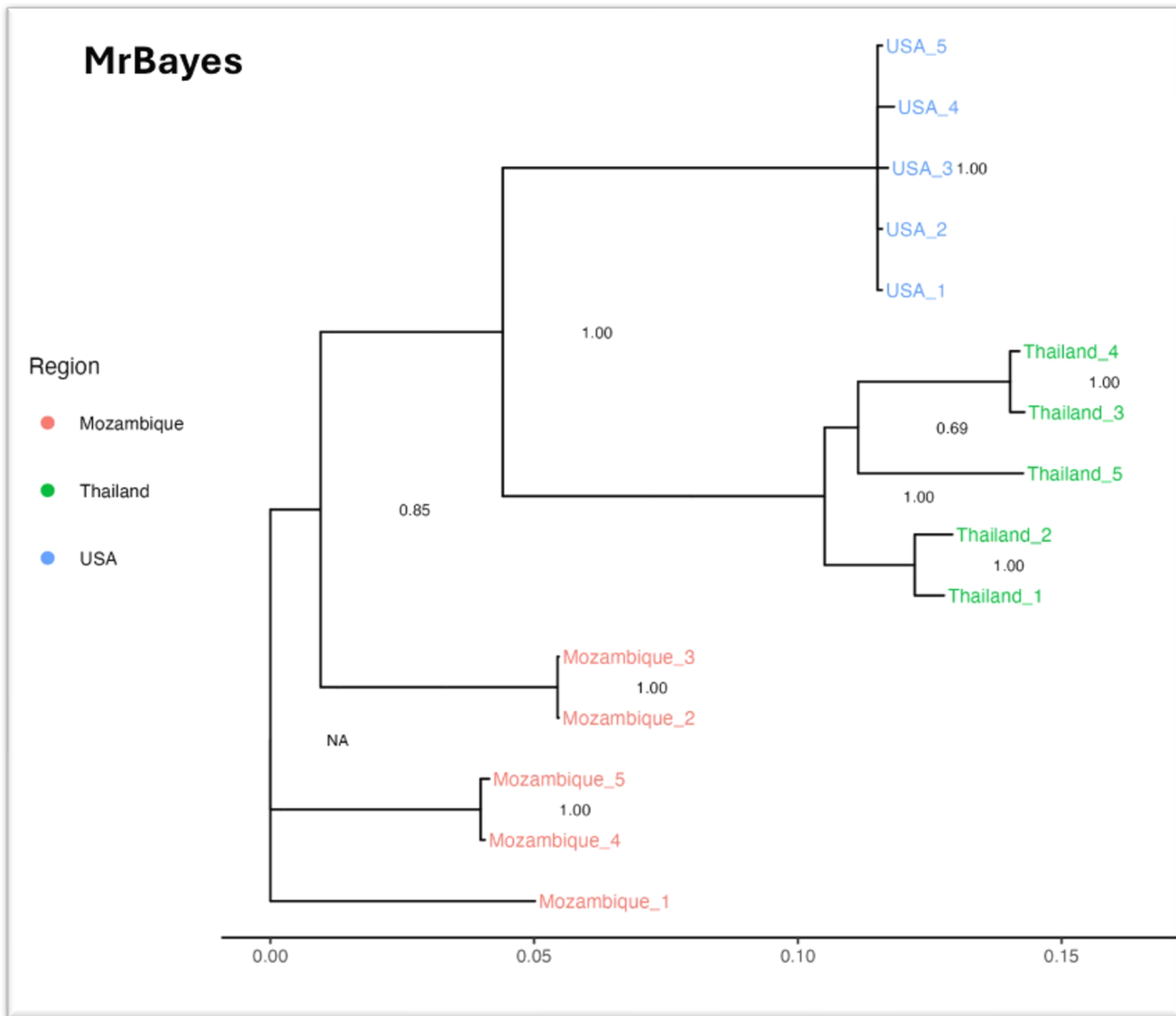


Figure 2. Bayesian inference tree constructed using MrBayes. Node values represent posterior probabilities, or the probability of each clade given the data and model parameters.

Inconsistent branch support values were seen in all 3 of the trees, with reliability varying greatly at each node. Additionally, a raw genetic distance model was used to create a distance matrix of all sequences compared pairwise to assess variability. Comparing two HIV-1 *pol* sequences from the same region produced much lower values than when comparing two from different regions, illustrating how sequences from the same region are similar to one another but distinct from other regions. Tracer analysis of BEAST log outputs produced variable, but mostly low effective sample size (ESS) values, implying poor support for estimated posterior distributions. Trace plots for posterior and likelihood showed a sharp increase at first with increased stability over time, though the plots never showed distinct convergence (Figure 3) (BEAST2 Developers, 2026; BEASTdoc, 2025).

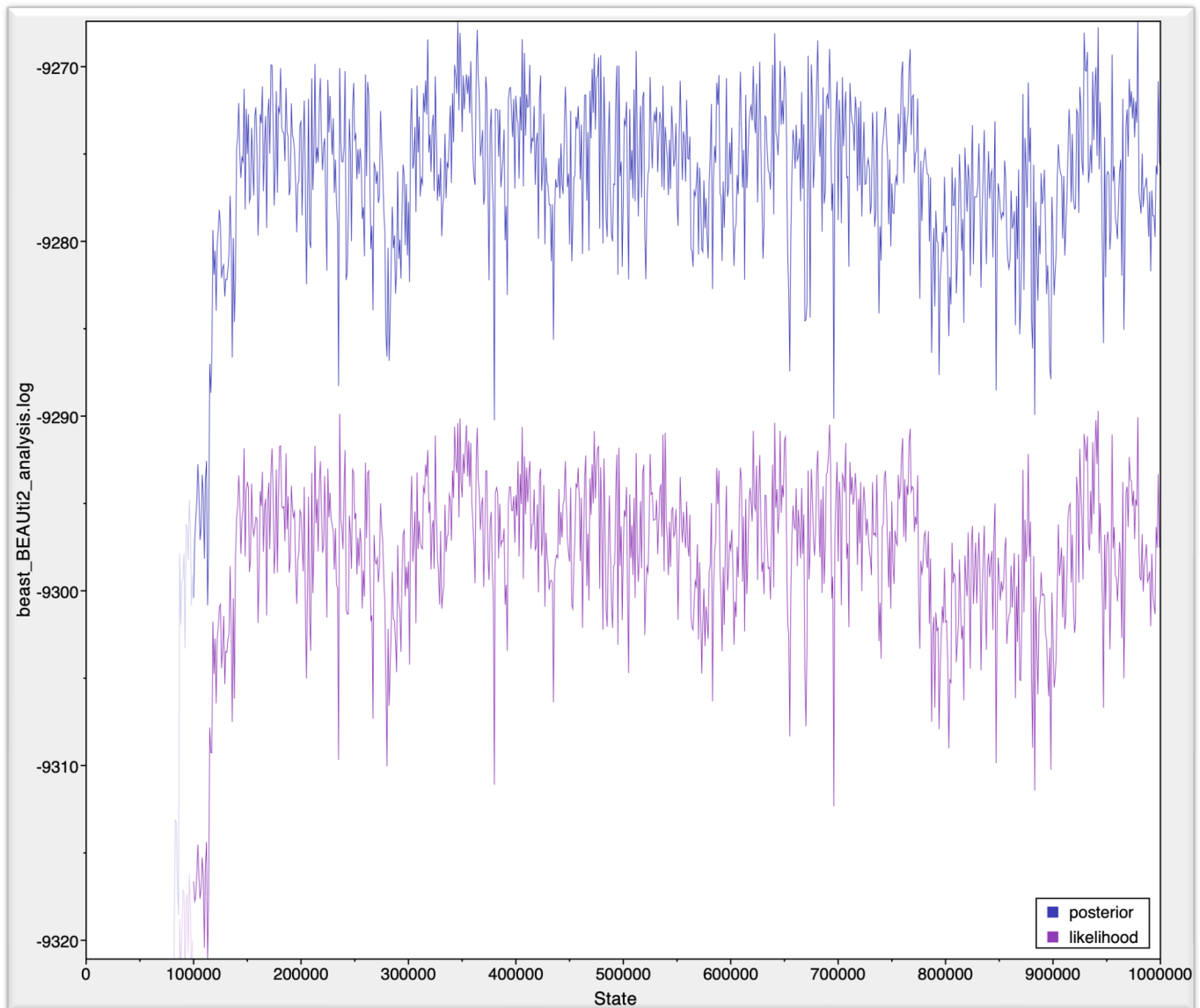


Figure 3. Combined Tracer plots of posterior (blue) and likelihood (purple) values. The x-axis shows MCMC generations, and the y-axis shows log values. The initial sharp increase around $x=100000$ represents the burn-in period prior to stabilization.

Discussion

Tree outputs show consistent and prominent clustering by country/region. Because each region has a different dominant subtype, what looks like clustering by geography may be partially attributed to subtype differences rather than regional variation. From a biological perspective, this implies a distinction between HIV-1 *pol* sequences regionally.

Looking at substructures formed by each country individually, variations in branch patterns suggest varying degrees of sequence diversity within regions. Sequences from the United States showed tight clustering, suggesting more conservation of *pol* genes on a sequence-to-sequence basis.

Mozambique and Thailand, however, form more than one subgroup of their own, suggesting more variety amongst sequences. The consistent patterns of clustering observed across all methods used

suggested a level of validity to our findings despite the low ESS and variable support values. Additionally, a high proportion of invariant sites (~76%) between the sequences was also consistent across models, supporting a semi-conserved nature of the *pol* gene across subtypes despite these low support values. These findings support the importance of continuing to study HIV-1 *pol* sequences, given their conserved nature across prominent regional subtypes, in order to aid in the development of new, globally-universal treatment methods.

A small sample size (n=15) was used for simplicity purposes while also learning new methods and software in class in order to write this paper. A mix of weak and strong phylogenetic signals across the project limits the ability to draw any clear conclusions, but does show important patterns that may encourage future work. Additionally, the lack of outgroup leaves no way to assess evolution over time, and the addition of one could be valuable to further understand the rise of treatment resistance across HIV-1 *pol* sequences.

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